

Pure Mathematics (40 marks)

- 1 When a solid turns into a gas without first becoming a liquid, the process is called sublimation. As a spherical mothball sublimes, its volume, in cm^3 , decreases at a rate, in cm^3 per day, proportional to its surface area.

Show that the radius of the mothball decreases at a constant rate. [3]

[It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

- 2 The curve C has equation

$$y = \frac{(ax+1)(4x+b)}{2x+3}, \quad x \in \mathbb{R}, \quad x \neq -\frac{3}{2},$$

where a and b are constants. It is given that $y = 2x - 1$ is an asymptote of C .

- (i) Find the values of a and b . [3]
- (ii) Sketch C , labelling all relevant point(s) and stating the equations of any asymptotes. [4]

- 3 The position vectors of the points U and V with respect to the origin O are $\mathbf{u}$ and $\mathbf{v}$ respectively, where $\mathbf{u}$ and $\mathbf{v}$ are non-zero and non-parallel vectors.

- (i) Show that $(\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u})$ can be written as $k(\mathbf{u} \times \mathbf{v})$, where k is a constant to be determined, justifying your working. [2]

The vectors $\mathbf{u}$ and $\mathbf{v}$ are now given by

$$\mathbf{u} = \mathbf{i} - \mathbf{j} + 2\mathbf{k} \quad \text{and} \quad \mathbf{v} = a\mathbf{i} + b\mathbf{j},$$

where a and b are constants.

- (ii) If $a = 2$ and $b = 3$, find a unit vector parallel to $(\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u})$. [2]
- (iii) If $\mathbf{v}$ is perpendicular to the vector $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, find $\mathbf{v}$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and b . Interpret the solution geometrically. [3]

- 4 A curve C has parametric equations

$$x = \cot \theta, \quad y = 2 \cos^2 \theta,$$

where $0 < \theta \leq \frac{1}{2}\pi$.

- (i) The point P on C has parameter $\theta = \frac{1}{6}\pi$. Show that the gradient of the tangent at P is $\frac{\sqrt{3}}{4}$. Hence find the equation of the normal, L , at P . [3]
- (ii) Sketch C and L on the same diagram, stating the coordinates of the points where the line L cuts the curve C and the x -axis respectively. [2]
- (iii) Find the exact area of the region bounded by the curve C , the positive y -axis and the line $y = \frac{3}{2}$. [3]

Not in the current syllabus
- (iv) Show that the Cartesian equation of C is given by $y = \frac{2x^2}{a + x^b}$, where a and b are constants to be determined. [2]
- (v) The region bounded by the curve C , the line L , the positive x -axis and the line $x = \frac{1}{\sqrt{3}}$ is rotated through 2π radians about the x -axis. Find the volume of the solid formed. [3]

[Turn over

5 A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_r = \begin{cases} 0 & \text{for } r = 2, \\ f(r-1) - 2f(r-3) & \text{for } r \text{ even, } r \neq 2, \\ f(r) & \text{for } r \text{ odd.} \end{cases}$$

- (i) Use the method of differences to find $\sum_{r=1}^{2n} u_r$. [3]

It is given that $f(r) = \ln\left(\frac{r+1}{r}\right)$.

Not in the current syllabus

- (ii) Use your answer to part (i) to show that $\sum_{r=1}^{2n} u_r = -\ln 2 + 2\ln\left(1 + \frac{1}{2n-1}\right)$. Hence state the value of the sum to infinity. [2]

- (iii) Find the smallest value of n for which $\sum_{r=1}^{2n} u_r$ is within 10^{-2} of the sum to infinity. [2]

- (iv) By considering the graph of $y = \frac{1}{x}$ for $x > 0$, show, with the aid of a sketch, that

$$\frac{1}{2n} < u_{2n-1} < \frac{1}{2n-1}, \quad n \in \mathbb{Z}^+. \quad [3]$$

Statistics (60 marks)

6 For events A and B it is given that $P(A) = 0.5$ and $P(B') = 0.35$.

- (a) Given that events A and B are not independent, find the range of values of $P(A \cup B)$. [2]

- (b) Given that $P(A|B') = 0.6$, find

(i) $P(A \cup B)$, [3]

(ii) $P(A' \cup B')$. [1]

- 7 The speed of the surface of a tennis court, affected by a variety of factors, including the physical makeup of the court, can be measured using the Court Pace Index (CPI). The CPI of a court varies with every match played on it. A higher mean CPI for a court indicates a faster court surface.

For the rest of this question, assume that data is collected under similar conditions.

At a tennis club, the CPI of the main court is a continuous random variable X . Based on past readings, the standard deviation of X is k and the expected value of X is 33.8.

- (i) A random sample of 50 readings is to be taken. When $k = 1.5$, estimate the probability that the mean value of X , for this sample, will be at most 0.5 units from 34. [2]

In January 2018, the main court was resurfaced with a new physical makeup.

- (ii) The club management claims that the new physical makeup of the resurfaced main court increases its mean CPI. A random sample of 60 readings were taken from the resurfaced main court after January 2018 and the mean CPI of this sample is found to be 34.3.

A test is carried out, at the 5% level of significance, to determine whether the resurfaced main court has a faster court surface.

Find the set of possible values of k for which the result of the test should be accepted. You should state your hypotheses clearly. [3]

- (iii) The club management later obtained information that the value of k is 0.8 and that X follows a normal distribution. Another random sample of 30 readings were taken from the resurfaced main court after January 2018 and the mean CPI of this new sample is found to be $\bar{x}$.

A test is carried out, at the 5% level of significance, to determine whether the mean CPI of the resurfaced main court has changed.

Find the set of possible values of $\bar{x}$ for which the result of the test should not be accepted. You should state your hypotheses clearly. [3]

[Turn over

- 8 In this question you should state clearly the values of the parameters of any normal distribution you use.

The masses in grams of apples have the distribution $N(80, 3^2)$ and the masses in grams of oranges have the distribution $N(100, 5^2)$.

- (i) Find the probability that the total mass of 2 randomly selected oranges is less than 205 grams. [2]
- (ii) Find the probability that the total mass of 4 randomly selected apples is more than three times the mass of 1 randomly selected orange. [3]

During the Community Health Week Event, visitors who take part in the activities are given gift boxes. Each gift box contains 3 apples and 2 oranges that are individually machine wrapped. The mass of wrapper used for each fruit is dependent on the mass of the fruit resulting in the mass of each wrapped fruit being 7% more than the mass of the fruit. The fruits are packed in a gift box and the mass of an empty gift box is normally distributed with mean 50 grams and standard deviation 2 grams. During packing, parts of the gift box is cut and removed. This process reduces the mass of each empty gift box by 10%.

- (iii) The probability that the total mass of a randomly selected gift box is less than k grams is 0.8. Find the value of k .

Find the probability that, at a particular collection point, the 25th gift box that is given is the 19th gift box whose total mass is less than k grams. [5]

- 9 The age of a species of trees can be determined by counting its rings, but that requires either cutting a tree down or extracting a sample from the tree's core. A forester attempts to find a relationship between a tree's age and its diameter instead. Based on past records, the forester found data for the diameter and the age (determined by the counting of its rings) of 8 trees of the same species that had been cut down. The results are given in the following table.

Diameter, D (inches)	1.8	6.6	9.9	10.8	12.8	13.2	15.4	16.5
Age, t (years)	5	8	16	12	22	28	30	40

- (i) Draw a scatter diagram for these values, labelling the axes clearly. [2]
- (ii) Find, correct to 4 decimal places, the product moment correlation coefficient between D and t . Comment on whether a linear model would be appropriate. [2]
- (iii) It was suggested that the tree's age can be modelled by one of the formulae

$$t = a + bD^2, \quad t = ae^{bD},$$

where a and b are constants.

Find, correct to 4 decimal places, the product moment correlation coefficient between

(A) t and D^2 ,

(B) $\ln t$ and D .

Explain which of $t = a + bD^2$ and $t = ae^{bD}$ best models the age of this species of trees based on the given data. [2]

- (iv) The forester wants to estimate the diameter of a tree given its age. Using a suitable regression line, find the required estimate of a tree that is 50 years old. Comment on the reliability of your estimate.

Explain why neither the regression line of t on D^2 nor the regression line of $\ln t$ on D should be used. [4]

[Turn over

10 Amy has four identical star shaped ornaments and six identical heart shaped ornaments.

- (a) Find the number of ways in which Amy can arrange the ten ornaments in a line. [1]

The star shaped ornaments are coloured Red, Green, Yellow and Purple, and the heart shaped ornaments are then coloured Red, Green, Yellow, Purple, Black and White.

- (b) Find the number of ways in which Amy can arrange the ten ornaments in a line such that exactly five heart shaped ornaments are next to each other. [3]
- (c) Amy then decides to arrange the ten ornaments in a circular manner. Find the number of ways in which all ornaments of the same colour are next to each other. [2]
- (d) Amy randomly distributes the ten ornaments into one group of four and two groups of three.
- (i) Show that the probability that the black and white ornaments are in the same group is $\frac{4}{15}$. [3]
- (ii) Find the probability that there are at least two heart shaped ornaments in the group of four given that the black and white ornaments are in the same group. [3]

- 11** The salad bar at a restaurant has n bowls each containing a different ingredient. Each customer to the restaurant is allowed only one visit to the salad bar. Jon visits the salad bar and makes a selection. At each bowl, he can take some or none of the contents, and he does not return to the bowl again. As the salad bar may not serve the same ingredients everyday, Jon does not know what ingredients will be served at each visit. On average, the probability that he takes some of the contents from each bowl is 0.4. For each visit to the salad bar, the number of different ingredients Jon takes is the random variable X .
- (i) State, in the context of this question, an assumption needed to model X by a binomial distribution, and explain why the assumption may not hold. [2]

Assume now that X has the distribution $B(n, 0.4)$.

- (ii) Jon visits the salad bar on a randomly chosen day. The probability that Jon takes exactly 7 different ingredients is denoted by P . Determine the value of n that gives the greatest value of P . [2]

Jon's selection on any day is independent of his selection on any other day. During a promotional week, the salad bar serves 6 different ingredients (A, B, C, D, E, F) on Monday and Tuesday, 6 different ingredients (G, H, I, J, K, L) on Wednesday and Thursday, and all 12 different ingredients (A to L) on Friday, Saturday and Sunday.

- (iii) If Jon visits the salad bar on Monday and Thursday of the promotional week, find the probability that he takes at least 3 different ingredients on each day. [2]
- (iv) If Jon visits the salad bar on Saturday of the promotional week, find the probability that he takes at least 6 different ingredients. [1]
- (v) Explain in context why the answer to part (iv) is greater than the answer to part (iii). [1]

After the promotional week, the restaurant decides to charge customers \$3 for a visit to the salad bar, and an additional \$2 if they take more than $\frac{n}{2}$ different ingredients. Kai visits the salad bar after the promotional week and makes a selection. The number of different ingredients Kai takes is the random variable K . It is given that

$$P(K \leq k) = \left(\frac{k}{n}\right)^2, \quad k = 0, 1, 2, \dots, n.$$

- (vi) If Kai visits the salad bar on a particular day after the promotional week when $n = 9$, find the expected amount Kai pays for that particular visit. [3]

Kai visits the salad bar on a randomly chosen day after the promotional week.

- (vii) Show that $P(K = k) = \frac{2k-1}{n^2}$, $k = 1, 2, \dots, n$. [1]

- (viii) Find, in terms of n , the expected number of different ingredients Kai takes on that day. [You may use the result $\sum_{r=1}^n r^2 = \frac{n}{6}(n+1)(2n+1)$.] [2]

***** End of Paper *****



RAFFLES INSTITUTION

2019 YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CLASS

19

MATHEMATICS

9758/01

PAPER 1

3 hours

Candidates answer on the Question Paper

Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the question paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved graphing calculator is expected, where appropriate.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

FOR EXAMINER'S USE					
Q1	Q2	Q3	Q4	Q5	Q6
Q7	Q8	Q9	Q10	Q11	Total

This document consists of 6 printed pages.

RAFFLES INSTITUTION
Mathematics Department

- 1 A curve C has equation

$$y = \frac{a}{x^3} + bx + c,$$

where a , b and c are constants. It is given that C has a stationary point $(-1.2, 6.6)$ and it also passes through the point $(2.1, -4.5)$.

- (i) Find the values of a , b and c , giving your answers correct to 1 decimal place. [4]
- (ii) One asymptote of C is the line with equation $x = 0$. Write down the equation of the other asymptote of C . [1]

- 2 Two variables u and v are connected by the equation $\frac{1}{u} + \frac{1}{v} = \frac{1}{20}$. Given that u and v both vary with time t , find an equation connecting $\frac{du}{dt}$, $\frac{dv}{dt}$, u and v . Given also that u is decreasing at a rate of 2 units per second, calculate the rate of increase of v when $u = 60$ units. [4]

3

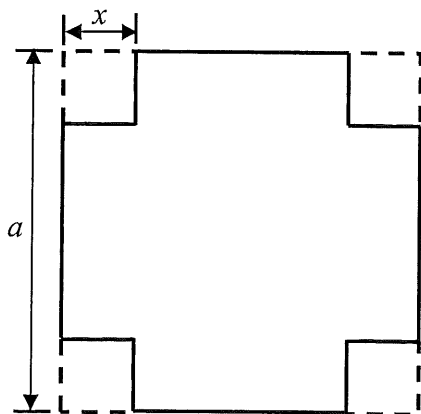


Fig. 1

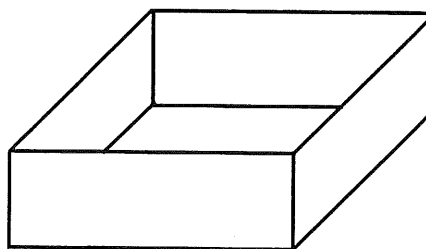


Fig. 2

Fig. 1 shows a square sheet of metal with side a cm. A square x cm by x cm is cut from each corner. The sides are then bent upwards to form an open box as shown in Fig. 2. Use differentiation to find, in terms of a , the maximum volume of the box, proving that it is a maximum. [6]